

AWS CLOUD TRAINING

Solution Architect Associate

Course Code - AWS SAA C03



CONTENT

Unit 1 - INTRODUCTION TO CLOUD COMPUTING

- Difference between traditional DC and cloud DC?
- What is virtualization?
- What is cloud and AWS (Amazon Web Services)?
- What is cloud computing?
- Deployments Model of Cloud Computing
- How to create an AWS Free Tier Account?
- AWS Global Infrastructure
- How to start with AWS?
- Manage your AWS Free Tier account.

Unit 2 - VPC (VIRTUAL PRIVATE CLOUD)

- Planning to setup your first virtual network?
- VPC Overview and Components
- Create your virtual VPC
- Difference Between IGW (Internet Gateway) and NAT Gateway
- Security Group
- Network Access Control List
- VPC Peering Connection
- VPC Endpoints
- Transit Gateway
- Elastic Network Interface Card
- VPC Pricing Overview

Unit 3 - STORAGE SERVICES - EBS (ELASTIC BLOCK STORAGE)

- Storage requirement and Techniques in DC
- Types of Storage Services in AWS
- What is EBS (Elastic Block Storage)?
- Types of Volume in EBS
- Difference between IOPS and Throughput
- Create EBS Volume for Linux and Windows Instances
- Detach EBS Volume on the Fly from one Instance and Attach with another Instance
- Modify the size and type of the EBS volume
- Share EBS volumes across AZ, region, and with other AWS accounts.
- EBS Pricing Overview

Unit 4: INSTANCE STORE

- Difference between EBS and Instance Store
- Use cases of Instance store
- Create your instance with an instance store disk
- Limitations of Instance store

Unit 5: EFS (ELASTIC FILE SYSTEM)

- EBS vs Instance Store vs EFS
- What is EFS (Elastic File System)
- Create a shared storage for instances.
- Access your share storage inside Instances
- Restrict the Client Access for EFS
- Pricing Overview

Unit 6: IAM (IDENTITY AND ACCESS MANAGEMENT)

- What is IAM?
- Components of IAM.
- Types of IAM Policies
- Write your custom IAM and inline policy.
- Create your custom IAM role
- Assume role
- Accessing AWS's other account resources

Unit 7: S3 (SIMPLE STORAGE SERVICE)

- Difference Between Block, File, and Object-Level Storage
- S3 overview and Components
- Create bucket and Performing some basic operations
- Access your bucket as a local drive
- S3 Features
- Server Access logging
- Static website hosting
- S3 VPC Endpoints
- Requester Pays
- Versioning
- Transfer Acceleration
- S3 Objects Life Cycle and Bucket replication
- Use cases of S3 storage classes
- Pricing Overview

Unit 8: S3 GLACIER

- S3 vs S3 Glacier
- S3 Glacier Overview and Components
- Create vault and upload the archives
- FastGlacier software uses

Unit 9: EC2 (ELASTIC COMPUTE CLOUD)

- EC2 Overview
- Instance backup using Snapshot
- Automate the backup with DLM
- Recover the key of Instance
- Create custom AMI
- Launch Template
- EC2 Instance Types
- Vertical Scalability
- Tenancy Model
- Service Models of EC2

Unit 10 - CLOUDWATCH

- Requirement of Monitoring tools
- CloudWatch Overview and Components
- Define the status Check (2/2)
- Checking the Metrics data
- Alarm definition
- Publish custom metrics by using CloudWatch Agent
- Push your App logs to CloudWatch LogGroup

Unit 11 - ELB (ELASTIC LOAD BALANCING)

- Load Balancer Requirement
- ELB Overview and Components
- Types of ELB
- Network Load Balancer
- Application Load Balancer
- Host-Based Load Balancing vs. Path-Based Load Balancing
- VPC Egress Routing
- Gateway Load Balancer
- Signing the public CA certificate in ACM
- Configure the HTTPS-based website

Unit 12 - ASG (AUTO SCALING GROUP)

- ASG Requirements with ELB
- ASG Overview and Components
- Perform Manual and Scheduled Horizontal Scalability
- Dynamic Scaling Policies
- Difference between Simple, Target Tracking, and Step Policies
- How to create ASG with ELB
- Updating EC2 Instances inside ASG

Unit 13 - ROUTE53

- DNS Requirement
- Route53 overview and Requirement
- Create private hosted zone
- Purchasing the public domain
- Create a public hosted zone
- ELB and S3 record creation in route53
- Health Check feature
- Route53 Routing Policy

Unit 14 - RDS (RELATIONAL DATABASE SERVICE)

- Database requirement
- RDS overview and requirement
- Types of Databases Supported in RDS
- Create your database and access your DB
- Taking backup and snapshot
- Restore your DB by using backup and snapshot
- Create read replica

Unit 15 - LAMBDA

- Requirement of PAAS
- Lambda Overview and requirement
- Creating Lambda Function and testing
- Adding the trigger (API Gateway and Event Bridge)
- Version Control in Function

Unit 16 - CLOUDFORMATION

- Requirement of IAC tool
- CloudFormation overview and Requirement
- Template Concept
- Create, update, and delete stack
- Creating Change Set

Unit 17 - CLI (COMMAND LINE INTERFACE)

- AWS CLI Requirement
- Installing AWS CLI in Windows and Linux
- Configuring the user profile
- Service provisioning through CLI

Unit 18 - SNS (SIMPLE NOTIFICATION SERVICE)

- SNS Overview and Requirement
- Create topic and add subscriptions
- Supported Subscribers
- SNS with SQS (Simple Queue Service)

Unit 19 - AWS VPN (VIRTUAL PRIVATE NETWORK)

- VPN requirement in network
- VPN overview
- Types of VPN in AWS
- OpenVPN Server
- How to import CA Certificate in ACM
- Client VPN Endpoints
- Site-to-Site VPN Connection

Unit 20 - CLOUD-MIGRATION

- AMS (Application Migration Service) Overview
- DMS (Database Migration Service) Overview
- Snowball Overview
- AWS DataSync Overview
- Storage Gateway Overview
- VM Import and Export Method
- Migrate your virtual server to AWS Cloud
- Perform post checks.

PROJECT CREATION AND EXAM TIPS

- Live project implementation
- Doing some feature testing
- Exam preparation tips

LEARNING OBJECTIVE OF THIS TRAINING

This AWS Cloud Computing course is designed to help you learn everything you need to know to become a skilled AWS Solutions Architect. You'll start with the basics of cloud and AWS, then step-by-step you'll learn how to build and manage secure networks, launch servers, store data, monitor resources, set up backups, use automation tools, and more. You'll work with key AWS services like EC2, S3, VPC, IAM, RDS, Lambda, CloudWatch, and Route53. By the end of the course, you'll be able to design and manage real-world cloud solutions on AWS and confidently prepare for the AWS Certified Solutions Architect Associate exam

PREREQUISITE FOR THIS TRAINING

- Student must have basic knowledge of Networking and Operating System (Windows/Linux)

CERTIFICATION PATH

After completion of the training, the candidate can attempt its official exam of the Amazon Web Services.

- Exam Code - AWS SAA C03
- Exam Name - AWS Certified Solutions Architect – Associate
- Exam Type - Multiple Choice or Multiple Response
- Exam Format - Online proctored
- Exam Location - Remote Location or Testing Center
- Number of Questions - 65 Questions
- Exam Duration - 130 minutes
- Maximum Marks - 1000
- Minimum Passing Score - 720
- Expiration - 3 Years

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